

Soft and Hard Contributions to Hard Pion Photoproduction

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Pion photoproduction at high transverse momentum supplements what can be learned in the standard probes of deep inelastic scattering and Drell-Yan processes. With polarized initial states there is sensitivity to the polarized gluon distribution, Δg , in leading order. This contrasts to other processes mentioned, which have no leading order gluon contribution. Additionally, in some kinematic regions the process occurs mainly due to pion production at short distances ("direct pion production," resulting in kinematically isolated pions), which gives sensitivity to the high- x valence quark distribution.

1. Meson Producing Processes

We are interested in using pion photoproduction, $(\gamma + A \rightarrow \pi + X)$ as a means of distinguishing between the competing models for the polarized quark and gluon distributions. There are three processes which are important in the kinematic range of interest. The two processes that are sensitive to the polarized quark and gluon distributions are shown in Fig. 1. At highest transverse momenta, k_t , of the outgoing pion the dominant process is producing the pion at short distance, via QCD, in what we call "direct pion production" [1]. This process is interesting because of its dependence on the polarized quark distributions. At moderate values of k_t the dominant process is a pion being produced due

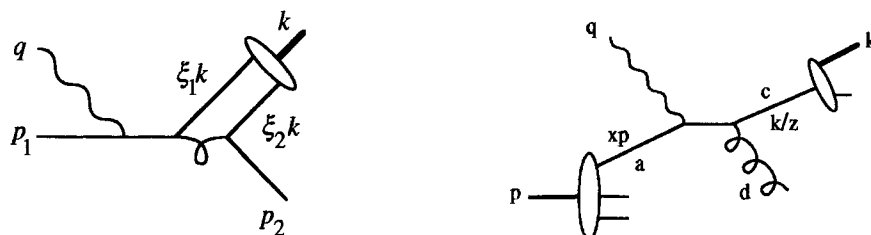


Figure 1. Direct pion production, left, and direct fragmentation production, right.

to fragmentation of a parton[2,3]. This process is of particular interest due to its leading order dependence on the polarized gluon distribution, Δg . This dependence is heightened by the large asymmetry, -100% , found in photon gluon fusion. In the kinematics that we are interested in, the contributions due to the resolved photon process turn out to be negligible.

The differential cross section for the direct process is

$$E_\pi \frac{d\Delta\sigma}{d^3k} = \frac{(s - m_N^2)x^2}{-\pi t} \sum_q \Delta G_{q/p}(x, \mu^2) \frac{d\Delta\hat{\sigma}(\gamma q \rightarrow \pi q')}{dt}. \quad (1)$$

$\Delta G_{q/p}$ is the polarized parton distribution function and the hatted differential cross section is for the subprocess, $(\gamma + q \rightarrow \pi + q)[1]$.

The differential cross section due to fragmentation is slightly different in that it involves a convolution integral over the momentum fraction z

$$E_\pi \frac{d\Delta\sigma}{d^3k} = \frac{s - m_N^2}{-\pi t} \sum_{a,c,d} \int_{z_{\min}}^1 \frac{dz}{z} x^2 \times \Delta G_{a/A}(x) \frac{d\Delta\hat{\sigma}}{dt}(\gamma + a \rightarrow c + d) D_{\pi/c}(z). \quad (2)$$

The $D_{\pi/c}$ is the fragmentation function for producing a pion from either a quark or gluon. The subprocess cross section, $(\gamma + a \rightarrow c + d)$, refers to either photon gluon fusion or compton scattering and may be found in Peralta *et al.*[2].

In order to accurately use the fragmentation process to determine ranges of Δg we need a good estimate of the background. There is justification for parameterizing the form of the cross section of photoproduction due to processes such as vector meson dominance, (VMD)[4], by using known couplings to relate the VMD cross section to proton-proton pion production

$$\begin{aligned} \omega_\pi \frac{d\sigma}{d^3k}(\gamma p \rightarrow \pi^+ X) = & \\ + \frac{\alpha}{\alpha_\rho^{eff}} \cdot 1.3 \cdot \frac{4}{3} \cdot \frac{2}{3} \cdot \omega_\pi \frac{d\sigma}{d^3k}(pp \rightarrow \pi^- X) \Big|_{90^\circ \text{CM}} g(\xi) & \\ + \text{non VMD contributions.} & \end{aligned} \quad (3)$$

This conversion is then just a function of the coupling constants and a scaling factor $g(\xi)$, where ξ is the scaled rapidity. Note that there is no polarization dependence given to the VMD cross section. This assumption agrees with a recent calculation of Manayenkov[5].

We can also use our formulas to calculate nearly on shell electroproduction by using the Weizäcker-Williams equivalent photon approximation and an appropriate photon number density[3], $N(E_\gamma)$.

$$d\sigma(eN \rightarrow \pi X) = \int_{E_{\min}}^{E_e} dE_\gamma N(E_\gamma) d\sigma(\gamma N \rightarrow \pi X). \quad (4)$$

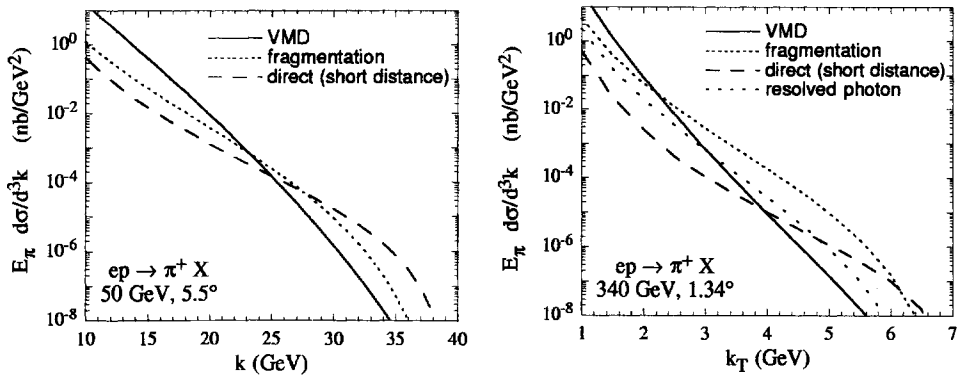


Figure 2. The invariant differential cross sections for $ep \rightarrow \pi^+ X$. The left figure is for incoming electron energy of 50 GeV, and pion lab angle of 5.5° such as is found at SLAC. The right figure is for incoming electron energy of 340 GeV and 1.34° which is equivalent to the proposed EPIC collider.

2. Results

Looking at energies typical of SLAC, we can see in Fig. 2 that there are regions of the kinematic spectrum that the direct process is dominant. At lower outgoing pion momentum the soft processes are found to dominate. Fig. 3 shows how our predictions of the polarization asymmetries for several competing polarized parton distributions compares with experimental data. The addition of the soft processes to previous calculations has had the effect of washing out some of the predicted asymmetries, destroying their sensitivity to polarized distributions. Unfortunately, this is the region that the currently available data primarily covers. As of yet, data points from the region of direct pion dominance are still unavailable.

Another interesting situation is exploring the kinematic region of a machine, EPIC, proposed by the University of Indiana that would have 4 GeV electrons colliding with 40 GeV protons. In these kinematics there is a large region where the fragmentation process is dominant, as seen in Fig. 2. This would give us a large region of sensitivity to the polarized gluon distribution. A measure of the polarization asymmetries, Fig. 4, would allow this sensitivity to be used to discriminate between competing models of Δg .

3. Summary

There are regions of hard pion photoproduction where pQCD calculable processes are dominant and interesting. Measuring polarization asymmetries in these regions can be an effective tool in determining Δg and Δq . We look forward to more data becoming available. It is also possible to extend this calculation to cover the kinematics of recent HERMES experiments with coincident pion pairs.

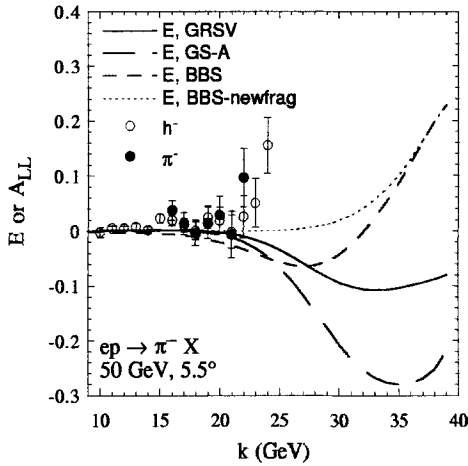


Figure 3. Polarization asymmetries for $ep \rightarrow \pi^- X$ with 50 GeV incident electrons and a pion lab angle of 5.5° . The data for charged hadrons and pions is from [6]. The curves are for several polarized distribution functions[7–9] and the newfrag refers to a parameterization of the fragmentation functions fit to recent HERMES data[10,4].

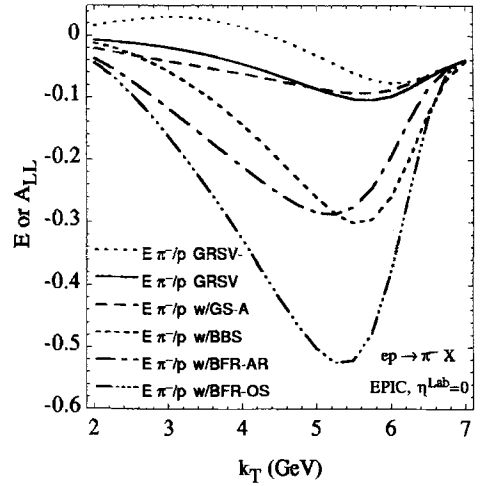


Figure 4. The polarization asymmetries for 340 GeV electrons scattering off a standing proton to produce a π^+ at 1.34° . The polarization asymmetries use the same polarized quark distribution, GRSV. There are five different models for Δg [11] and one curve ("GRSV-") with Δg set to zero.

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